

01 & 02 Assignment – Docker, Docker Hub and Containerization

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1. L1 - Create Docker file and build the docker container Application Image for the application build in Jenkins Module.

- Launch EC2 Instance and Connect Using MobaXterm.

Instance summary for i-09b11d304f24b0fc5 (Docker Server) Info

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Instance ID i-09b11d304f24b0fc5	Public IPv4 address 51.20.137.166 open address	Private IPv4 addresses 172.31.15.198
IPv6 address -	Instance state Running	Public DNS ec2-51-20-137-166.eu-north-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-15-198.eu-north-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-15-198.eu-north-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 51.20.137.166 [Public IP]	VPC ID vpc-04fbb5dfae5f66a87	Auto Scaling Group name -
IAM Role -	Subnet ID subnet-0c6c6ee27b50c1eda	

- Install Docker and Git on EC2

```
[root@ip-172-31-15-198 ec2-user]# docker --version
Docker version 25.0.14, build 0bab007
[root@ip-172-31-15-198 ec2-user]#
```

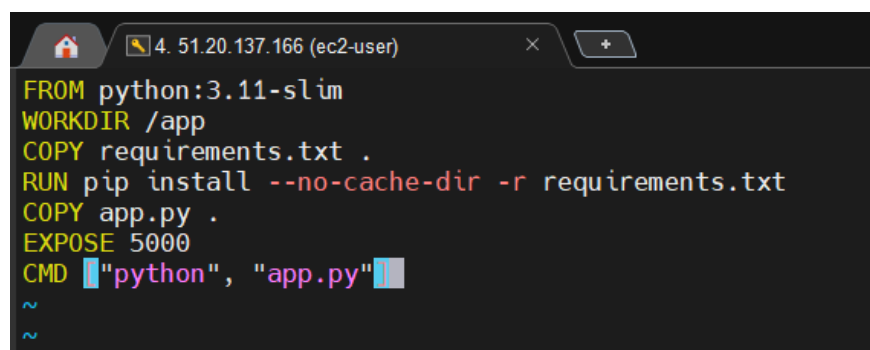
```
[root@ip-172-31-15-198 ec2-user]# git --version
git version 2.50.1
[root@ip-172-31-15-198 ec2-user]#
```

- Clone Project from GitHub to EC2

```
[root@ip-172-31-15-198 ec2-user]# git --version
git version 2.50.1
[root@ip-172-31-15-198 ec2-user]# git clone https://github.com/PoojaPrakash08/Docker_Assignment.git
Cloning into 'Docker_Assignment'...
remote: Enumerating objects: 9, done.
remote: Counting objects: 100% (9/9), done.
remote: Compressing objects: 100% (6/6), done.
remote: Total 9 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (9/9), done.
[root@ip-172-31-15-198 ec2-user]# ls
Docker_Assignment ECR-ECS-Project Kastro-Ecom-DC lb-project
```

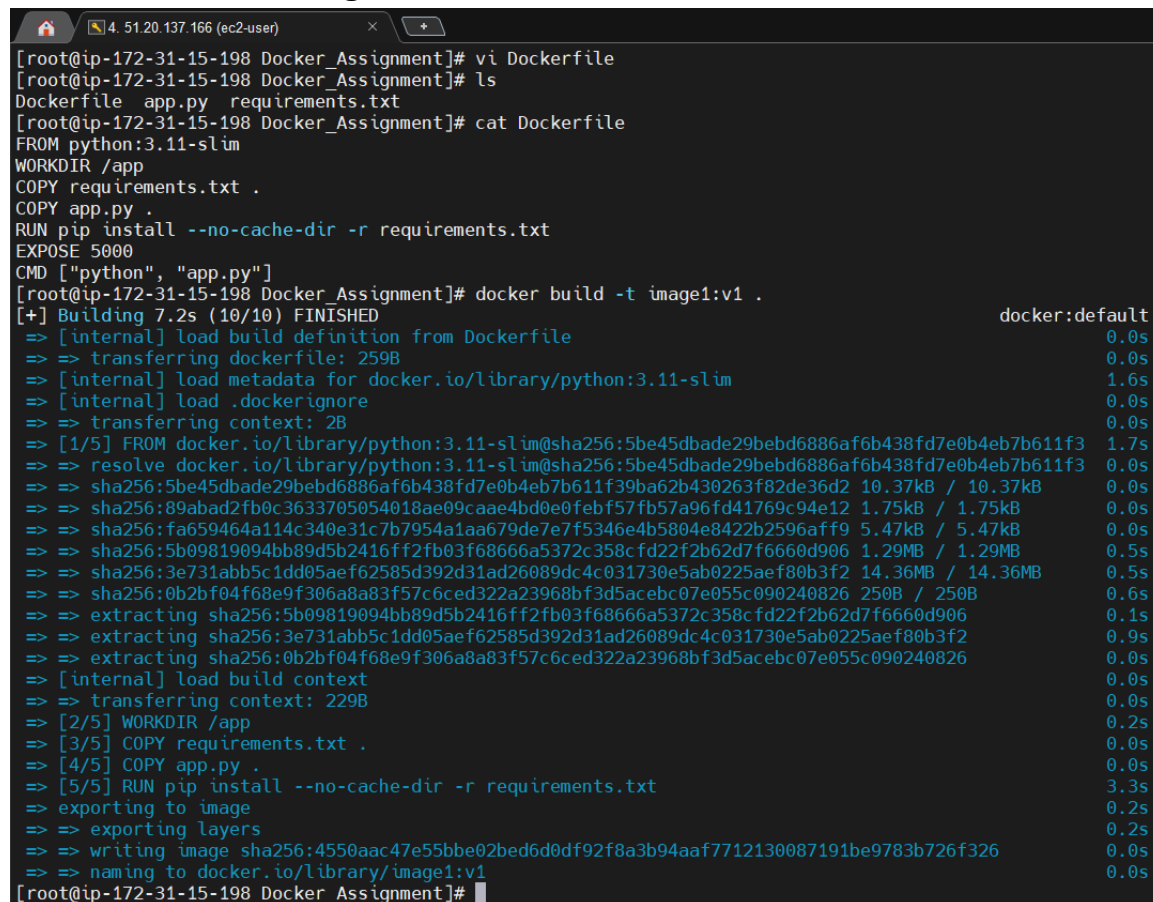
- Navigate to Project and Create Dockerfile

```
[root@ip-172-31-15-198 ec2-user]# ls
Docker_Assignment ECR-ECS-Project Kastro-Ecom-DC lb-project
[root@ip-172-31-15-198 ec2-user]# cd Docker_Assignment
[root@ip-172-31-15-198 Docker_Assignment]# ls
app.py requirements.txt
[root@ip-172-31-15-198 Docker_Assignment]#
```



A terminal window titled '4. 51.20.137.166 (ec2-user)' displays the content of a Dockerfile. The file starts with 'FROM python:3.11-slim', followed by 'WORKDIR /app', 'COPY requirements.txt .', 'RUN pip install --no-cache-dir -r requirements.txt', 'COPY app.py .', 'EXPOSE 5000', and 'CMD ["python", "app.py"]'. The window has a dark background with light-colored text.

- Build Docker Image and Run Container

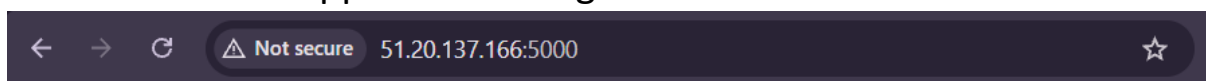


A terminal window titled '4. 51.20.137.166 (ec2-user)' shows the process of building a Docker image. It starts with 'vi Dockerfile' and 'ls' in the 'Docker_Assignment' directory. Then, 'cat Dockerfile' shows the same content as the previous screenshot. Finally, 'docker build -t image1:v1 .' is executed, resulting in a long output of progress messages and a final 'FINISHED' status. The output includes details about the build definition, context transfer, layer extraction, and image export. The window has a dark background with light-colored text.

2. L2 - Create the Container using the same Application Image and run the application in a Web Browser using container port mapping

```
[root@ip-172-31-15-198 Docker_Assignment]# docker images
REPOSITORY    TAG       IMAGE ID       CREATED        SIZE
image01        v1        bd44b5614161   12 seconds ago 140MB
<none>         <none>    2d4389d8910f   11 minutes ago 140MB
[root@ip-172-31-15-198 Docker_Assignment]# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS        NAMES
386dd5fec959   image01:v1     "python app.py"         5 seconds ago Up 4 seconds  0.0.0.0:5000->5000/tcp, :::5000->5000/tcp   my-python-container
[root@ip-172-31-15-198 Docker_Assignment]#
```

- Access the Application Using Public IP



Hello from Docker!